Rajat Saxena

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About

Interdisciplinary background in computational and theoretical studies across astrophysics and statistical mechanics. Interested in the physics of emergent phenomena, machine learning, complex and intelligent systems. Additionally, keen on working on optimisation and variational problems in physics. Passionate about science communication, philosophy, and history.

Education

- **MSc in Physics** — *October 2024 - Present*
  Ludwig Maximilian University (LMU) of Munich; ***Grade: 2.24*** *(1 Highest, 5 Fail)* — Munich, Germany
  Advisor: Dr Steffen Rulands, Professor, LMU

- **BSc Blended Physics** — *June 2024*
  Savitribai Phule Pune University; ***CGPA: 9.32*** *(10 Highest, 4 Fail)* — Pune, India
  Advisor: Dr Deepak Dhar, Distinguished Professor Emeritus, IISER - Pune

Research Experience

- **Arnold Sommerfeld Centre for Theoretical Physics, LMU Munich** — *October 2025 - Present*
  Master Thesis
  Mentor: Prof. Dr Steffen Rulands, Onurcan Bektas
  - Investigating the role of specialisation in multi-agent systems as a form of emergent collective memory to address the plasticity-memory dilemma
  - Assembling and programming autonomous robots to perform collective tasks. Implementing multiple neural network architectures trained on multimodal sensory inputs.
  - Designing and simulating experiments to model and analyse collective behaviour and performance outcomes under different tasks.

- **Universitäts-Sternwarte München (University Observatory Munich)** — *March 2025 - October 2025*
  Research Intern
  Mentor: Dr Tilman Birnstiel, Professor
  - Developed N-body simulations to model the gravitational collapse of a dust cloud in the presence of a central star and a pre-formed gas disk using REBOUND. Available: []
  - Simulated particle trajectories integrated with gas dynamics during protostar formation, leveraging hydrodynamic snapshots from (from Tārā) to interpolate density, velocity, and temperature fields in both time and space at particle locations. Available: []
  - Implemented semi-implicit and exponential midpoint integration schemes in Python and performed numerical simulations on a self-built codebase, achieving faster execution than REBOUND. Available: []
  - Summarised the group's research for Astrobites, translating complex results into accessible articles. Available: []

- **Indian Institute of Science Education and Research (IISER) – Pune** — *August 2023 - May 2024*
  Bachelor Thesis
  Mentor: Dr Deepak Dhar, Distinguished Professor Emeritus
  - Studied foundational statistical mechanics and phase transitions through standard texts to build a strong theoretical base.
  - Developed a Python simulation of a 1D zero-temperature random field Ising model to investigate discrete magnetisation jumps in ferromagnetic materials with impurities and defects.
  - Analysed the impact of lattice size, random field standard deviation, and probability distributions on magnetisation curves, quantifying parameter-dependent behaviour.

- **Inter-University Centre for Astronomy and Astrophysics (IUCAA)** — *June 2023 - July 2023*
  Summer Intern
  Mentor: Mr Ashish Mhaske, Scientific and Technical Officer
  - Led a student team to measure 21-cm neutral hydrogen power spectrum across different Galactic longitudes using a conical horn antenna.

- By analysing Doppler-shifted HI line profiles, we derived the Milky Way's rotation curve and observed the discrepancy between the theoretically predicted and the observed rotation curves.
- Designed and implemented a Python data-analysis pipeline to process source and calibration data.

- **National Centre for Radio Astrophysics (NCRA) - TIFR** *September 2022 – April 2023*
  Research Intern
  Mentor: Dr Yashwant Gupta, Distinguished Professor and Centre Director
  - Developed Python code for folding and binning time-series data to generate pulsar profiles for the Crab Pulsar using uGMRT data.
  - Implemented C routines to define on/off gates, enabling precise separation of pulsar on-pulse and off-pulse emission intervals.
  - Applied pulsar timing, dedispersion techniques, and correlator analysis, leveraging software tools such as TEMPO2 and GMRT Pulsar Tool (gptool).

AI/ML Projects

- **CNNs for Mass Estimation of X-Rays Clusters**
  Employed Convolutional Neural Networks (CNNs) to estimate Galaxy cluster masses from simulated X-ray images and redshift data. The redshift values were concatenated to the convoluted and flattened image data before being fed to the fully connected network. Available: [🌐]
- **Explicit Time-depedent Neural Quantum States**
  Developed a Transformer encoder–decoder architecture to represent time-dependent spin configurations and predict $\psi(s, t)$, with a physics-informed loss function minimising deviations from the Schrödinger equation. Available: [🌐]
- **Emergent Weight Morphologies in Deep Neural Networks**
  Investigated the emergent structural patterns in neural network weights that arise independently of the training data. Also examined how these structural patterns change with alterations in hyperparameters and training strategies.
- **Neural Networks for Quantum Many-Body Physics**
  Implemented a Restricted Boltzmann Machine (RBM) ansatz within a Monte Carlo framework to approximate the ground state of a Rydberg atom chain. Utilised the expressive capacity of RBMs to represent many-body wavefunctions and employed stochastic gradient optimisation to minimise the variational energy.
- **Simulation-Based Inference for Bayesian Cosmological Data Analysis**
  Developed a simulation-based inference framework using autoencoder compression and normalising flows to recover cosmological parameters from Gaussian random fields. Benchmarked learned posteriors against analytic Fourier-space likelihoods, identifying effects of compression loss and autoencoder overfitting.

Outreach Experience

- **LIGO – India Education and Public Outreach (LI-EPO), IUCAA** *January 2023 – July 2024*
  Intern
  Mentor: Dr Debarati Chatterjee, Associate Professor and Chair LI-EPO
  - Developed outreach content and social media posts to communicate LIGO's science, and maintained the interferometer demonstrations for public engagement.
  - Conducted awareness campaigns across Indian colleges, promoting the upcoming observatory.
  - Designed and led an introductory Python workshop for undergraduates, and assisted in organising Astronomy Teacher Training Workshops and local outreach at Hingoli (LIGO India site).
- **Science Popularisation Centre (SciPop), IUCAA**
  Volunteer
  - Operated telescopes and led stargazing sessions for school students and the general public, and conducted training for new volunteers in telescope operation.

Mentorship Experience

- **Teaching Assistant at Quest Classes, Pune** (Aug–Oct 2024): Taught 12th-grade Physics (optics) and 11th-grade Mathematics (inequalities, combinatorics, binomial theorem).
- **Co-founder & Research Head, IDSS Space Society** (Oct 2021 – Jul 2023): Organised guest lectures, stargazing nights, school outreach programs, and space-themed MUNs. Recognised as a "Registered Space Tutor of ISRO" by CBPO, ISRO.

Selected Conferences

Oral Presentations

- **10th IMPRS-IS Interview Symposium**, Max Planck Institute for Intelligent Systems *February 2026*
- **30th Young Scientists' Conference on Astronomy and Space Physics**, National University of Kyiv *April 2024*
- **10th National Student Symposium on Physics**, Indian Association Physics Teachers (IAPT) *October 2023*

Poster Presentations

- **Symposium on Magnetism and Spintronics**, Indian Institute of Technology - Bombay *July 2024*
- **National Space Science Symposium**, Indian Space Research Organization (ISRO) *February 2024*

Relevant Workshops

- **Trento-Innsbruck Quantum Information Tour (TIQT)**, AISF *Feb 2026*
- **German Italian Physics Exchange**, German Physical Society & Associazione Italiana Studenti di Fisica *Sept 2025*
- **Summer University for Plasma Physics and Fusion Research**, Max Planck Institute for Plasma Physics *Sept 2025*
- **Summer School in Theoretical (Astro)Physics**, St Xavier's College & IUCAA *June 2024*
- **Course on Nuclear Experiments**, HBCSE-TIFR *March 2024*
- **Workshop on Data Science in Astronomy**, IUCAA *December 2023*

Skills

- **Programming Languages:** Python, Mathematica, Arduino programming (C/C++)
- **Python Libraries:** Numpy, Scipy, Astropy, REBOUND, Matplotlib, Pandas
- **ML Libraries:** Keras/TensorFlow, Sklearn, OpenCV
- **Software and Tools:** LAMMPS, ORCA, Astrometrica, LaTeX, Git, WordPress, Hugo
- **Telescope Handling:** Dobsonian & Newtonian telescopes (EQ/Altazimuth mounts)
- **Languages:** English (IELTS: 7.5), Hindi (Native), Marathi (Professional)

Awards and Achievements

- **Exceptional Accomplishments in Physics and Science Communication** *March 2024*
  Interdisciplinary School of Science, Savitribai Phule Pune University
  - Conferred by the Hon. Vice Chancellor in recognition of outstanding academic performance and science communication initiatives.
  - Acknowledged as a significant departmental honor highlighting excellence in both research and outreach.
- **Telescope Building Grant (INR 60,000)** *2022–23*
  Savitribai Phule Pune University
  - Received university funding to design and construct a Newtonian telescope.

Professional Memberships

- **Student Member of *Deutsche Physikalische Gesellschaft (German Physical Society)*** *2025 - Present*
- **Student Member of *Associazione Italiana Studenti di Fisica*** *2025 - 2026*
- **Student Member of *Indian Physics Association*** *2024 - Present*

Publications

[1] **Saxena, Rajat.** (2024). Introduction to Statistical Mechanics and Simulation of Random Field Ising Model. 10.13140/RG.2.2.34379.81443. Available: [🌐]

[2] **Saxena, Rajat** & Padalkar, Gauri & Patil, Yash & Sherkar, Shreenath & Satam, Harsh & Shah, Manasvi & Rajopadhye, Chinmayee. (2024). Galaxy Rotation Curve Measurements using a Horn Antenna. Available: [🌐]